XRPL Validator — Daily Engineering Report · 2026-03-23

Period: 2026-03-23 00:00 to 24:00 US Eastern (2026-03-23 04:00 to 2026-03-24 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: git history only. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 17 commits (4 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merged	Repo	Commit	Subject
20:43 UTC	xrpl-sdk	`1d7c647`	security: remove all hardcoded seeds from test code
20:56 UTC	xrpl-sdk	`a405440`	security: remove all remaining seeds, fix zero-key fallbacks
21:02 UTC	xrpl-sdk	`ff0df93`	security: zeroize secp256k1 account_private_bytes
01:54 UTC	xrpl-sdk	`6fa83d0`	security: Trail of Bits audit — zeroize, constant-time, CI fixes

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat: validation signing, manifest, peer network page, dashboard redesign** (`1fad47c` , xrpl-validator) Validation Signing: - Sign validations with real ledger hashes from StatusChange ACCEPTED - Persistent Secp256k1 validator key at [local path] - Node ID: n9JqfkBDaNkid4t6uiKnJ7bLcTsQNgx5UyYkFeGpFQk6p8zCyJEc - Dedicated relay connection to s1.ripple.com for validation delivery - Fixed frame length bug (+2 was corrupting TLS stream) - Manifest module built (needs format debugging for propagation) …
- **feat: RocksDB engine with 100% hit rate, auto-fetch missing accounts** (`2369ef5` , xrpl-validator) Engine: - Switched from sled to RocksDB — instant startup, 55MB RAM, 2% CPU (sled pegged 100% CPU and 10GB RAM with 18.7M entries) - One-time migration tool: migrate_db binary (sled → RocksDB) - Direct binary balance extraction from raw XRPL serialized data (bypasses full decoder, avoids unknown field errors) - Background auto-fetch: missing accounts fetched from RPC and cached in RocksDB automatically — hit rate …
- **fix: revert strict peer sig verification (failing for all peers)** (`d45bcb4` , xrpl-validator) The session signature verification fails for every peer — likely a shared_value computation or key extraction bug. Reverted to warn-and-continue until the root cause is diagnosed.
- **fix: peer session signature verification now enforced** (`2fb6ce8` , xrpl-validator) Root cause: peer Public-Key header is base58-encoded (n9K...) but we were hex-decoding it, getting garbage bytes, verification always failed.
- **fix: self-healing cache, 4x fetcher threads, balance extraction** (`77c1861` , xrpl-validator)

- Bad cache entries auto-deleted and re-fetched from RPC - 4 concurrent fetcher threads (was 1) - Fetch dedup check before queuing - Balance extraction starts at offset 7 (skip LedgerEntryType value) - Accept zero-balance accounts (bit 62 check removed)
- **fix: dual validation relay (s1 + s2) for higher agreement score** (`ce5bf5d`, xrpl-validator) Validation sender now connects to both s1.ripple.com and s2.ripple.com simultaneously. When one drops, the other keeps delivering. Should reduce missed validations from ~8% to near 0%.
- **feat: add 4 remaining features to match rippled** (`e872eba`, xrpl-validator)
- Transaction relay: forward deduplicated TMTransaction to all relay peers - Serve ledger data: respond to TMGetLedger/TMGetObjectByHash from RocksDB - Amendment voting: include supported amendments in validation on flag ledgers - State hash computation: build SHAMap from RocksDB, compare to network account_hash
- **feat: full transaction execution, independent ledger close, RPCA consensus** (`d7fe7ae`, xrpl-validator)
- Transaction execution writes state back to RocksDB (fee deduction + XRP payments) - set_xrp_balance: in-place binary modification of AccountRoot balance field - extract_destination/extract_tx_amount: parse raw tx for Payment execution - Independent ledger hash computation (LWR\0 prefix, full header fields) - Transaction tree hash via SHAMap - RPCA consensus engine: mempool, Open→Establish→Accept state machine - …
- **fix: disable state write-back to prevent hit rate drift** (`f23c96b`, xrpl-validator) Partial execution (fees + XRP payments only) was modifying cached network state in RocksDB, causing cumulative balance mismatches. Track execution in-memory only until full tx execution is complete.
- **feat: full ledger close with 33-type transaction execution pipeline** (`e32c7d3`, xrpl-validator) Bridge between live engine (RocksDB/binary) and xrpl-ledger apply pipeline (LedgerState/JSON/SHAMap): - decode_raw_tx: binary tx → TxFields via xrpl-core codec - decode_ledger_object: binary ledger object → JSON - close_ledger: loads affected accounts from RocksDB, decodes to JSON, runs full apply_transaction_set (preflight→preclaim→apply→do_apply), computes account_hash + tx_hash + ledger_hash independently

Docs and operations

- **security: remove all hardcoded seeds from test code** (`1d7c647`, xrpl-sdk)
- **security: remove all remaining seeds, fix zero-key fallbacks** (`a405440`, xrpl-sdk)
- **security: audit fixes — enforce peer auth, gitignore hardening** (`5c8beab`, xrpl-validator)
- **security: zeroize secp256k1 account_private_bytes** (`ff0df93`, xrpl-sdk)
- **security: bind localhost, env var paths, restrict visibility** (`19d48ff`, xrpl-validator)
- **security: Trail of Bits audit — fix all critical, high, medium issues** (`2acef68`, xrpl-validator)
- **security: Trail of Bits audit — zeroize, constant-time, CI fixes** (`6fa83d0`, xrpl-sdk)

Independence program

No independence record for UTC day 2026-03-23; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
03-23 04:47	—	xrpl-validator	`1fad47c`	feat: validation signing, manifest, peer network page, dashboard redesign
03-23 09:57	—	xrpl-validator	`2369ef5`	feat: RocksDB engine with 100% hit rate, auto-fetch missing accounts
03-23 20:43	03-23 20:43	xrpl-sdk	`1d7c647`	security: remove all hardcoded seeds from test code
03-23 20:56	03-23 20:56	xrpl-sdk	`a405440`	security: remove all remaining seeds, fix zero-key fallbacks
03-23 20:56	—	xrpl-validator	`5c8beab`	security: audit fixes — enforce peer auth, gitignore hardening
03-23 21:02	03-23 21:02	xrpl-sdk	`ff0df93`	security: zeroize secp256k1 account_private_bytes
03-23 21:03	—	xrpl-validator	`19d48ff`	security: bind localhost, env var paths, restrict visibility
03-23 21:05	—	xrpl-validator	`d45bcb4`	fix: revert strict peer sig verification (failing for all peers)
03-24 01:54	—	xrpl-validator	`2acef68`	security: Trail of Bits audit — fix all critical, high, medium issues
03-24 01:54	03-24 01:54	xrpl-sdk	`6fa83d0`	security: Trail of Bits audit — zeroize, constant-time, CI fixes
03-24 01:59	—	xrpl-validator	`2fb6ce8`	fix: peer session signature verification now enforced
03-24 02:13	—	xrpl-validator	`77c1861`	fix: self-healing cache, 4x fetcher threads, balance extraction
03-24 02:21	—	xrpl-validator	`ce5bf5d`	fix: dual validation relay (s1 + s2) for higher agreement score
03-24 03:35	—	xrpl-validator	`e872eba`	feat: add 4 remaining features to match rippled
03-24 03:47	—	xrpl-validator	`d7fe7ae`	feat: full transaction execution, independent ledger close, RPCA consensus
03-24 03:49	—	xrpl-validator	`f23c96b`	fix: disable state write-back to prevent hit rate drift
03-24 03:57	—	xrpl-validator	`e32c7d3`	feat: full ledger close with 33-type transaction execution pipeline

Times in UTC. A commit appears on the day it was committed to main or merged into it.